

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 5: Rules of the Air & SERA

Syllabus Source: CAE Oxford (pp. 101-156) | Pass Target: >= 90% AviationExam

1. REGULATORY AUTHORITY & PIC RESPONSIBILITIES (SERA)

Standard European Rules of the Air (SERA) are established by Commission Implementing Regulation (EU) No 923/2012 and ICAO Annex 2. The Pilot-in-Command (PIC) has final authority over the disposition of the aircraft and is responsible for compliance with the rules of the air, but may deviate in an emergency in the interest of safety.

2. VISUAL METEOROLOGICAL CONDITIONS (VMC) MINIMA TABLE

VMC minima define when a flight may proceed under Visual Flight Rules (VFR). AviationExam questions strictly test exact cloud distance and visibility numbers across airspace classes:

Altitude / Level Band	Airspace Classes	Min. Flight Visibility	Distance from Clouds
FL 100 and above	Classes A, B, C, D, E, F, G	8 km	1,500 m horizontal, 1,000 ft (300 m) vertical.
Below FL 100 (or above 3,000 ft AMSL whichever is higher)	Classes B, C, D, E, F, G (Class A: IFR only)	5 km	1,500 m horizontal, 1,000 ft (300 m) vertical.
At or below 3,000 ft AMSL (or 1,000 ft AGL whichever is higher)	Classes F and G	5 km*	Clear of clouds and with the surface in sight. (*Reduced to 1,500 m if speed <= 140 kt IAS).

[!] EXAM TRAP (AVIATIONEXAM) - VFR in Class A Airspace & Special VFR (CTR)

1. Airspace Class A: VFR flights are STRICTLY PROHIBITED (IFR only). 2. Special VFR (Control Zone CTR only): Granted by ATC when weather is below VMC: - Ground visibility: at least 1,500 m (800 m for helicopters); - Ceiling: at least 600 ft (180 m); - Clear of clouds and in sight of the surface; - Maximum indicated airspeed: 140 kt IAS.

3. MINIMUM SAFE HEIGHTS FOR VFR FLIGHTS

- Congested Areas (Cities, Towns, Open-Air Gatherings): At least 1,000 ft (300 m) above the highest obstacle within a 600 m radius of the aircraft, and high enough to permit safe emergency landing.
- Other Areas (Rural / Sea): At least 500 ft (150 m) above ground or water, or 500 ft above the highest obstacle within 150 m of the aircraft.

4. SEMICIRCULAR CRUISING LEVELS (SERA.5015)

Cruising levels are based on MAGNETIC TRACK. In RVSM airspace (FL 290 to FL 410), vertical separation is 1,000 ft:

Magnetic Track	IFR Cruising Levels	VFR Cruising Levels (Above 3,000 ft)
Track 000° to 179° (EAST)	ODD Flight Levels (FL 050, 070, 090, 110, 130...)	ODD + 500 ft (FL 055, 075, 095, 115...)

**Track 180° to 359°
(WEST)**

EVEN Flight Levels (FL 060, 080, 100,
120, 140...)

EVEN + 500 ft (FL 065, 085, 105,
125...)

5. RIGHT-OF-WAY RULES (COLLISION AVOIDANCE)

- Hierarchy: Balloons > Gliders > Airships > Power-driven aircraft. (Aircraft towing another has right of way over power-driven aircraft).
- Converging: When two aircraft converge at approximately the same level, the aircraft having the other on its RIGHT shall give way.
- Approaching Head-on: Each aircraft shall alter its heading to the RIGHT.
- Overtaking: The aircraft being overtaken has right-of-way. The overtaking aircraft keeps clear by altering heading to the RIGHT.
- Landing: Aircraft in flight or on ground shall give way to aircraft landing or in final approach. Between two approaching to land, the LOWER aircraft has right of way, but shall not cut in.

6. AIRCRAFT LIGHTS (SERA.3215)

Navigation lights must be displayed from SUNSET TO SUNRISE (and any other period prescribed by authority): - Port Wing Light: RED light showing an unbroken arc of 110° from dead ahead to port. - Starboard Wing Light: GREEN light showing an unbroken arc of 110° from dead ahead to starboard. - Tail Light: WHITE light showing an unbroken arc of 140° visible aft. - Anti-Collision Lights: Flashing red or white light displayed by day and night to indicate operating engines.

7. ATC LIGHT GUN SIGNALS (TOWER TO AIRCRAFT)

Light Signal	Meaning to Aircraft IN FLIGHT	Meaning to Aircraft ON GROUND
Steady Green	Cleared to land	Cleared for take-off
Flashing Green	Return for landing (followed by steady green)	Cleared to taxi
Steady Red	Give way to other aircraft and continue circling	STOP immediately
Flashing Red	Aerodrome unsafe, DO NOT LAND	Taxi clear of runway in use
Flashing White	Land at this aerodrome and proceed to apron	Return to starting point on aerodrome
Red Pyrotechnic	Notwithstanding previous instructions, DO NOT LAND for the time being	--

8. VISUAL GROUND SIGNALS IN AERODROME SIGNAL AREA

Ground Signal Symbol	Official Meaning & Requirement (Annex 14 / SERA)
Red square with yellow diagonals	Landings prohibited; prohibition is likely to be prolonged.
Red square with one yellow diagonal	Special precautions landing: state of maneuvering area poor, land with care.
White dumb-bell	Landings, take-offs, and taxiing must be confined to paved/hard surfaces.
White dumb-bell with black bars	Take-offs and landings on runways only, but other movements not confined to paved taxiways.
White cross on runway/taxiway	Runway or taxiway unserviceable; closed to all aircraft movements.

Landing T (White or Orange)	Landings and take-offs shall be in a direction parallel to the shaft toward the crossarm.
Tetrahedron (White or Orange)	Indicates landing direction: aircraft lands in direction towards which apex points.
Arrow on runway (Yellow/White)	Preceding area before displaced threshold fit for taxi and take-off, but NOT for landing.

9. MARSHALLING SIGNALS (ANNEX 2 / SERA.3301)

Signal Action	Marshaller Visual Description & Crew Action
Wingman / Identify Gate	Raise right arm fully vertical, wand pointing up; left wand pointed horizontally toward gate.
Straight Ahead	Bend arms repeatedly upward and backward from elbows with wands pointing up.
Turn Left	Point left arm and wand down; move right arm repeatedly upward/backward.
Turn Right	Point right arm and wand down; move left arm repeatedly upward/backward.
Normal Stop	Cross arms and wands overhead rapidly with wands crossed.
Emergency Stop	Abruptly cross arms overhead and wave wands vigorously side to side.
Insert Chocks	Arms extended palms inwards; swing arms from outwards to cross below waist.
Cut Engines	Extend arm with wand at throat level; draw wand across throat in slicing motion.

10. INTERCEPTION PROCEDURES (ANNEX 2 / SERA)

[!] EXAM TRAP (AVIATIONEXAM) - Interception of Civil Aircraft (AviationExam Core)

- Interception is a measure of last resort. - Intercepted aircraft must immediately: 1) Follow instructions given by intercepting aircraft; 2) Notify ATS unit immediately; 3) Squawk Mode A Code 7700 (unless instructed otherwise); 4) Monitor emergency frequency 121.5 MHz (and 243.0 MHz). - Interceptor Day Signals: - Rocking wings and slow left turn: 'You have been intercepted, follow me'. Response: Rocking wings ('Understood, will comply'). - Abrupt breakaway climb and turn: 'You may proceed'.

11. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What does a flashing red light from the aerodrome control tower indicate to an aircraft in flight?

- [A]: Cleared to land
- [B]: Aerodrome unsafe, DO NOT LAND
- [C]: Give way to other aircraft and continue circling
- [D]: Return for landing

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Flashing red in flight means aerodrome unsafe, do not land (steady red means give way and continue circling).

Q2: In Class G airspace at 2,000 ft AMSL, what are the VMC minima for an aeroplane flying at 130 kt IAS?

- [A]: 8 km visibility and 1,500 m horizontal from clouds

- [B]: 5 km visibility, 1,500 m horizontal, 1,000 ft vertical
- [C]: 1,500 m visibility, clear of clouds and in sight of the surface
- [D]: Special VFR clearance required

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. At or below 3,000 ft AMSL (or 1,000 ft AGL) in Class F/G, VMC visibility is 5 km, but may be reduced to 1,500 m if speed is ≤ 140 kt, clear of clouds and surface in sight.

Q3: What visual aerodrome signal indicates that landings are prohibited?

- [A]: A red square with a single yellow diagonal
- [B]: A red square with two yellow diagonals
- [C]: A white dumb-bell
- [D]: A landing T

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A red square with yellow cross diagonals signifies that landings are prohibited and prohibition is likely prolonged.

Q4: Two power-driven aircraft are converging at approximately the same altitude. Which aircraft has the right of way?

- [A]: The faster aircraft
- [B]: The aircraft that has the other on its right
- [C]: The aircraft that has the other on its left
- [D]: The heavier aircraft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. The aircraft which has the other on its right shall give way; therefore, the aircraft that has the other on its left has right of way.

Q5: What are the minimum weather conditions for a Special VFR flight in a CTR?

- [A]: Ground visibility $\geq 1,500$ m, ceiling ≥ 600 ft, clear of clouds, ground in sight
- [B]: Visibility ≥ 5 km, clear of clouds
- [C]: Visibility ≥ 800 m and ceiling $\geq 1,000$ ft
- [D]: Any condition provided IFR traffic is absent

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Under SERA, Special VFR requires ground visibility $\geq 1,500$ m (800 m helo) and cloud ceiling ≥ 600 ft.